

Lecture 3

Floer theory with support

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1 From the first two lectures to support

In Lecture 1 we constructed Floer operations on surfaces by counting bigons and polygons. We computed the wrapped endomorphisms of a cotangent fiber in T^*S^1 and used this algebra to explain the affine HMS statement. Lecture 2 introduced the geometric settings, coefficient rings, and Lagrangian torus fibrations that we want to use in higher dimensions.

Today we explain how to restrict Floer theory to a subset of the ambient symplectic manifold. The objects will still be Lagrangian branes in the *whole* manifold. It is their morphism complexes that depend on the support. This will let us compute a global category from local categories, and then construct a mirror by gluing local endomorphism algebras.

The basic construction is

$$\text{acceleration data} \longrightarrow \text{telescope} \longrightarrow \text{completion}.$$

We first explain this in Morse theory. We then recall the analytic Floer background needed to replace Morse trajectories by strips and polygons. The last part concerns the canonical A_∞ structures, descent, and the torus example.

The open-string construction and descent statements are from the preprint with Mohammed Abouzaid and Yoel Groman [1]; the closed-string operadic construction is in [2]. We use the same brane and grading conventions as in the first two lectures. A Hamiltonian sign convention used for the support construction is specified in Section 3.

For both exercises: You may consult the literature, but the writeup, calculations, and pictures must be your own. You may suppress local systems. To avoid orientation signs, you may work over characteristic two.

2 Morse theory with support

2.1 The indicator function and acceleration data

Let X be a closed smooth manifold and $K \subset X$ a compact subset. The function we would like to study is

$$\chi_K(x) = \begin{cases} 0, & x \in K, \\ +\infty, & x \notin K. \end{cases}$$

Here $+\infty$ is a prescription for a limiting process. Choose Morse functions $f_i : X \rightarrow \mathbb{R}$ such that

$$f_i < f_{i+1}, \quad f_i|_K < 0, \quad f_i(x) \longrightarrow \chi_K(x).$$

Together with Morse–Smale gradient-like vector fields and monotone continuation data, these form an *acceleration datum* for K .

Use Morse cochains: a critical point has degree equal to its Morse index, and the differential counts upward trajectories. On $CM^*(f_i; \mathbb{Z})$ put

$$\text{val} \left(\sum_x a_x x \right) = \min_{a_x \neq 0} f_i(x), \quad \text{val}(0) = +\infty.$$

The differential does not decrease this valuation. A continuation trajectory for a family f_s satisfies

$$\frac{d}{ds} f_s(\gamma(s)) = \partial_s f_s(\gamma(s)) + df_s(X_s)(\gamma(s)) \geq 0,$$

provided $\partial_s f_s \geq 0$ and X_s is gradient-like for f_s . Consequently the continuation maps form a valuation non-decreasing ray

$$CM^*(f_1) \xrightarrow{\kappa_1} CM^*(f_2) \xrightarrow{\kappa_2} CM^*(f_3) \longrightarrow \cdots.$$

Each map is an ordinary quasi-isomorphism. The filtration is where the support is recorded.

2.2 The telescope and its completion

For any ray of cochain complexes $C_1 \rightarrow C_2 \rightarrow \cdots$, set $S = \bigoplus_{i \geq 1} C_i$, and let $\kappa : S \rightarrow S$ be the shift induced by the continuation maps. One model for its homotopy colimit is

$$\text{Tel}(C_i) = \text{Cone}(\text{id} - \kappa : S \longrightarrow S).$$

More explicitly, introduce a symbol q of degree -1 and write

$$\text{Tel}(C_i) = S \oplus qS, \quad D(x) = dx, \quad D(qx) = x - \kappa x - q dx.$$

Give qx the same valuation as x , and use the minimum valuation on a finite sum. The completed telescope is the degreewise completion:

$$CM_X^*(K) := \widehat{\text{Tel}}(CM^*(f_i)).$$

A series is allowed in this completion precisely when, below each fixed valuation bound, only finitely many of its summands occur.

Here is the elementary mechanism that makes completion matter. Suppose x_i are cycles with $\kappa_i(x_i) = x_{i+1}$ and $\text{val}(x_i) \rightarrow +\infty$. In the completion,

$$D \left(\sum_{i \geq 1} qx_i \right) = \sum_{i \geq 1} (x_i - x_{i+1}) = x_1.$$

The infinite sum converges, so this class dies. If the valuations of the x_i stay bounded, this particular primitive is not available. Thus classes pushed to arbitrarily high valuation disappear, whereas classes remaining near K can survive.

Without completion the cohomology is $H^*(X; \mathbb{Z})$, since all continuation maps induce isomorphisms. Completing the *chain-level* telescope is what changes the answer.

2.3 Restriction, invariance, and the topological answer

Two acceleration data can be interleaved after passing to subsequences. Monotone continuation maps in both directions, together with their homotopies, give the usual sandwiching argument for independence of the result. If $K' \subset K$, then $\chi_K \leq \chi_{K'}$, so there is a restriction map

$$CM_X^*(K) \longrightarrow CM_X^*(K').$$

On models built from individual sequences these maps compose up to specified homotopies. A model incorporating all auxiliary choices makes restriction strictly functorial.

The topological result recalled in the slides, due to Özel, is

$$H^*(CM_X^*(K)) \cong \check{H}^*(K; \mathbb{Z}),$$

where the right side is Čech cohomology of K with its subspace topology. For example, a closed hemisphere in S^2 gives $\mathbb{Z}$ in degree zero and zero in the other degrees. For a manifold or a sufficiently regular compact subset, this agrees with the familiar cohomology of K . Thus “with support on K ” here does not mean the ordinary relative cohomology $H^*(X, X \setminus K)$.

Morse theory satisfies descent for finite compact covers. For $K = K_1 \cup K_2$ the sequence is encoded by the homotopy pullback

$$CM_X^*(K) \simeq \text{Cone} \left(CM_X^*(K_1) \oplus CM_X^*(K_2) \xrightarrow{\text{res}_1 - \text{res}_2} CM_X^*(K_1 \cap K_2) \right) [-1].$$

At the level of approximating functions, union corresponds to minimum and intersection to maximum. After suitable tangentialization and smoothing, the associated continuation square is homotopy Cartesian. This is the pattern we will use in Floer theory, where there is an additional Poisson-commuting condition. The closed-string descent construction is developed in [3].

3 The Lagrangian Floer background

3.1 Geometric assumptions and coefficients

There are two settings in the lecture. In the exact setting, (M, λ) is a complete finite-type Liouville manifold and $\omega = d\lambda$. Its branes are exact embedded Lagrangians, compact or conical at infinity, with chosen primitives $\lambda|_{L_i} = df_i$. We use contact-type Floer data at infinity so that the relevant solutions satisfy a C^0 bound. Exactness excludes nonconstant holomorphic disks and spheres.

In the closed setting, fix the grading structure of Lecture 2 and use closed embedded graded branes with the orientation and local-system data from that lecture. We work in the tautologically unobstructed setting: for the chosen brane almost complex structures there are no nonconstant holomorphic disks. The classical construction in [1] assumes spherical nonnegativity,

$$\omega(A) > 0 \implies \langle c_1(TM), A \rangle \geq 0 \quad (A \in \pi_2(M)).$$

In our closed graded examples $2c_1(TM) = 0$, so this condition is automatic. The absence of disks is still a separate condition on the objects. The preprint handles a more general weakly unobstructed setting with a fixed disk potential; we stay in the zero-potential case here.

Use the Novikov field $\Lambda = \Lambda_{\mathbb{Q}}$ of Lecture 1 and its valuation ring

$$\Lambda_{\geq 0} = \{a \in \Lambda : \text{val}(a) \geq 0\}, \quad \Lambda_{\geq E} = \{a \in \Lambda : \text{val}(a) \geq E\}.$$

The supported construction will first be made over $\Lambda_{\geq 0}$ and completed there. We extend scalars to Λ afterwards. In the exact setting it is also possible to work over $\mathbb{Z}$, using the action filtration directly. All statements involving signs use the chosen brane orientations; characteristic two gives the unsigned version.

3.2 Hamiltonian chords and Floer cochains

For this lecture we use the convention

$$\iota_{X_{H_t}} \omega = dH_t. \quad (1)$$

This is the convention of the support calculation.

Let $H : [0, 1] \times M \rightarrow \mathbb{R}$. A chord from L_0 to L_1 is a path

$$\dot{x}(t) = X_{H_t}(x(t)), \quad x(0) \in L_0, \quad x(1) \in L_1.$$

Equivalently, evaluate a chord at its starting point to identify

$$\mathcal{X}_H(L_0, L_1) \cong L_0 \cap (\phi_H^1)^{-1}(L_1), \quad x \mapsto x(0).$$

Thus we can keep L_0 fixed and move the second brane by the inverse Hamiltonian isotopy. For an autonomous Hamiltonian with positive radial slope on a Liouville end, this inverse flow wraps L_1 positively. This compares our convention with the positive wrapping in Lecture 1 by wrapping the second brane.

Nondegeneracy means that $\phi_H^1(L_0)$ is transverse to L_1 at the corresponding endpoint. In particular, $L_0 = L_1$ is allowed: it is the Hamiltonian-perturbed pair that must be transverse. For regular data,

$$CF^*(L_0, L_1; H) = \bigoplus_x \Lambda_{\geq 0} x.$$

Each chord carries an integer degree, obtained from the transported gradings just as for intersections in Lecture 1. More precisely, a summand also carries its orientation line and $\text{Hom}(E_0|_{x(0)}, E_1|_{x(1)})$ when local systems are present. We suppress these factors in the formulas.

3.3 Floer strips and the differential

Choose a compatible almost complex structure J_t . The equation is

$$\partial_s u + J_t(u)(\partial_t u - X_{H_t}(u)) = 0, \quad u(s, 0) \in L_0, \quad u(s, 1) \in L_1, \quad (2)$$

with limits x_- and x_+ as $s \rightarrow -\infty$ and $+\infty$. We regard x_- as the input and x_+ as the output. A regular moduli space has dimension $|x_+| - |x_-|$ before quotienting by translation in s . Isolated solutions modulo translation therefore contribute to a differential of degree $+1$.

For a solution of (2), put

$$E^{\text{top}}(u) = \int_{\mathbb{R} \times [0, 1]} u^* \omega + \int_0^1 H_t(x_+(t)) dt - \int_0^1 H_t(x_-(t)) dt.$$

For fixed H this equals the nonnegative geometric energy. The Novikov-weighted differential is

$$dx_- = \sum_{x_+} \sum_{[u] \text{ rigid}} \epsilon(u) T^{E^{\text{top}}(u)} x_+.$$

The signs and local-system parallel transports generalize those in Lecture 1. When $H = 0$, the exponent is the symplectic area, so the formula reduces to the bigon count already studied on surfaces.

For regular data, the ends of one-dimensional moduli spaces are broken strips. The compactness and unobstructedness assumptions exclude the additional bubbling that could contribute to this boundary. Counting its oriented endpoints gives $d^2 = 0$. This is the analytic version of the picture from Exercise 1 of Lecture 1. For details of the classical transversality argument under the above assumptions, see [1, Sections 3–5]; [6] gives background on the usual Floer and Fukaya constructions.

3.4 Action and the continuation energy identity

In the exact setting the action is single-valued:

$$\mathcal{A}_H(x) = \int_0^1 x^* \lambda + \int_0^1 H_t(x(t)) dt + f_0(x(0)) - f_1(x(1)). \quad (3)$$

With the metric $g(v, w) = \omega(v, Jw)$, the Floer equation is upward gradient flow for this action. In a closed manifold we use the energy of a strip directly; a single-valued action would require reference paths and cappings.

Now let (H_s, J_s) interpolate between (H_-, J_-) and (H_+, J_+) , constant for $|s|$ large. Replace H_t, J_t in (2) by $H_{s,t}, J_{s,t}$. A direct calculation gives

$$\begin{aligned} E^{\text{top}}(u) &:= \int u^* \omega + \int_0^1 H_+(x_+(t)) dt - \int_0^1 H_-(x_-(t)) dt \\ &= E^{\text{geo}}(u) + \int_{\mathbb{R} \times [0,1]} (\partial_s H_s)(t, u(s, t)) ds dt. \end{aligned} \quad (4)$$

The derivative $\partial_s H_s$ here differentiates the Hamiltonian family at a fixed point of M . In the exact case Stokes' theorem also gives

$$E^{\text{top}}(u) = \mathcal{A}_{H_+}(x_+) - \mathcal{A}_{H_-}(x_-).$$

If $\partial_s H_s \geq 0$, the exponent in (4) is nonnegative. Counting rigid continuation strips therefore defines a map over $\Lambda_{\geq 0}$,

$$\kappa_{-+} : CF^*(L_0, L_1; H_-) \longrightarrow CF^*(L_0, L_1; H_+),$$

which does not decrease valuation.

The boundary of a one-dimensional continuation moduli space gives $d\kappa = \kappa d$. A one-parameter family of continuation data gives a degree -1 chain homotopy, and higher parameter spaces provide higher coherence. These are precisely the additional maps needed when we use many Hamiltonians and compare different acceleration sequences. For closed M continuation is a quasi-isomorphism after passing to Λ ; the required inverse need not preserve the $\Lambda_{\geq 0}$ lattice. This distinction is essential for support.

3.5 Polygons and the A_∞ relations

For branes $L_0, \dots, L_k$, take a disk with k incoming boundary punctures and one outgoing boundary puncture. Label its boundary arcs by the L_i , prescribe Hamiltonian chords at the punctures, and choose compatible Hamiltonian and almost complex data on the disk, with input Hamiltonians $H_1, \dots, H_k$ and output Hamiltonian H_{out} . Counting rigid solutions gives

$$\mu^k : CF^*(L_{k-1}, L_k; H_k) \otimes \dots \otimes CF^*(L_0, L_1; H_1) \longrightarrow CF^*(L_0, L_k; H_{\text{out}})[2 - k].$$

4 Floer complexes with support

4.1 The completed acceleration model

Let M be closed and let $K \subset M$ be compact. Fix two branes L_0, L_1 . Choose nondegenerate Hamiltonians H_i and regular monotone continuation data such that

$$H_i < H_{i+1}, \quad H_i|_{[0,1] \times K} < 0, \quad H_i(t, x) \longrightarrow \chi_K(x).$$

This is a Floer acceleration datum. Define its supported cochain model by

$$CF_{M,\text{acc}}^*(L_0, L_1; K; \Lambda_{\geq 0}) = \widehat{\text{Tel}}_i CF^*(L_0, L_1; H_i; \Lambda_{\geq 0}). \quad (5)$$

The completion is the degreewise Novikov completion:

$$\widehat{\text{Tel}}_i CF_i = \varprojlim_{E \rightarrow +\infty} (\text{Tel}_i CF_i) / \Lambda_{\geq E} (\text{Tel}_i CF_i).$$

We then set

$$HF_M^*(L_0, L_1; K; \Lambda) = H^*(CF_{M,\text{acc}}^*(L_0, L_1; K; \Lambda_{\geq 0}) \otimes_{\Lambda_{\geq 0}} \Lambda).$$

Localization is flat, so one may instead take cohomology of the completed complex first and then tensor with Λ .

The order of operations matters: retain the Novikov lattice, take the telescope, complete, and only then invert T . Since T^E is a unit in Λ , completion cannot be defined by the ideals $T^E \Lambda$. One can equivalently formulate the construction using filtered Λ -spaces and valuation non-decreasing maps.

This also explains the difference from Lecture 1. Ordinary direct limits of Floer *cohomology* were sufficient for the wrapped calculation there. Here we need the filtered chain complexes and their continuation maps before taking cohomology. Exercise 2 will exhibit nonzero supported classes in winding sectors whose finite-stage Floer cohomology over Λ vanishes.

4.2 Action filtration in the exact setting

For exact branes we can instead count strips over $\mathbb{Z}$ without recording the Novikov weight, and give a chord the valuation $\mathcal{A}_H(x)$. Equation (4) implies that the differential and monotone continuation are valuation non-decreasing. Complete the resulting telescope with respect to this action filtration.

The comparison between the two conventions is useful. If x_N is a generator of valuation zero in the Novikov-weighted complex, then, after adjoining Novikov coefficients, its unweighted, action-filtered version corresponds to

$$x_{\text{act}} = T^{\mathcal{A}_H(x)} x_N.$$

Indeed, a strip contributes $T^{\mathcal{A}_H(y) - \mathcal{A}_H(x)} y_N$ to dx_N , so it contributes just $\pm y_{\text{act}}$ to dx_{act} . This calculation also applies to continuation maps with different endpoint Hamiltonians.

4.3 What changes when the support changes?

The branes themselves are always branes in M . They need not be contained in K , and they are not replaced by intersections $L_i \cap K$ in the definition. The notation $\text{Fuk}_M(K)$ keeps track of this ambient manifold dependence.

An inclusion $K' \subset K$ gives a restriction map

$$CF_M^*(L_0, L_1; K) \longrightarrow CF_M^*(L_0, L_1; K').$$

It is constructed by comparing acceleration data for $\chi_K \leq \chi_{K'}$. As in Morse theory, sandwiching proves invariance, with continuation homotopies providing compatibility of comparisons. The canonical model in the next section packages these maps into strict restriction functors.

After passing to Λ , support on M recovers ordinary Floer cohomology. Support on the empty set gives zero. More generally, if one of the underlying branes is disjoint from K , the supported morphism complex is acyclic [1, Proposition 2.3 and Remark 2.4]. At the opposite extreme, if one brane is contained in K , restriction from the ambient manifold to K preserves its Floer morphisms.

5 Keeping all the choices: a canonical model

5.1 Why a telescope is not the whole construction

An acceleration datum is an efficient way to compute the complex for one pair (L_0, L_1) . To multiply two such complexes, however, we must also choose Floer data on a three-punctured disc and a sufficiently large output Hamiltonian. Associativity requires homotopies between these choices; compatibility of those homotopies requires further homotopies. The same issue arises when comparing restriction maps for several nested supports. A chosen sequence and its continuation maps do not, by themselves, specify this additional data.

5.2 Operads and the Floer multicategory

A non-symmetric dg operad $\mathcal{O}$ consists of chain complexes $\mathcal{O}(k)$ of k -input operations, with a unit and associative substitution maps

$$\mathcal{O}(k) \otimes \mathcal{O}(i_1) \otimes \cdots \otimes \mathcal{O}(i_k) \longrightarrow \mathcal{O}(i_1 + \cdots + i_k).$$

An algebra over it is a complex C with compatible maps $\mathcal{O}(k) \otimes C^{\otimes k} \rightarrow C$. For example, chains on the moduli spaces of stable discs with ordered boundary punctures form such an operad. A fundamental chain of the space of discs with k inputs has cohomological degree $2 - k$; its boundary expresses the A_∞ relation familiar from Lecture 1.

A *multicategory*, or colored operad, allows different input and output types, called colors. For a collection $\mathcal{B}$ of branes, use ordered pairs (L, L') as colors. The operation space

$$\mathcal{O}_{\mathcal{B}}((L_{k-1}, L_k), \dots, (L_0, L_1); (L_0, L_k)) = \mathcal{O}(k)$$

records composable strings; other operation spaces are zero. An algebra over this multicategory is therefore a collection of morphism complexes with operations parametrized by discs.

The Floer multicategory $\mathcal{H}_K$ has finer colors

$$Y = (L_0, (H, J), L_1), \quad H|_{[0,1] \times K} < 0,$$

with nondegenerate Hamiltonian chords and the required regularity. Its multimorphism complexes are cubical chains on spaces of regular, monotone Floer data on possibly broken punctured discs, with the specified input and output data. A zero-dimensional chain is a choice of data, a one-dimensional chain is a family interpolating between choices, and so

on. Composition concatenates the corresponding configurations. There are two compatible multifunctors

$$\mathrm{CF} : \mathcal{H}_K \longrightarrow \mathrm{Ch}_{\Lambda_{\geq 0}}, \quad \pi : \mathcal{H}_K \longrightarrow \mathcal{O}_{\mathcal{B}}.$$

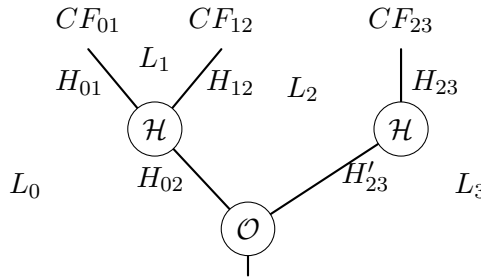
The first assigns a Floer complex to a color and counts solutions for each family of data. The second forgets the Hamiltonian and almost complex structure and remembers the underlying stable disc and its brane labels. We use an uncurved collection of branes as above.

We now extend CF along π . The relevant extension is the *operadic homotopy left Kan extension*: it combines all these complexes and operations, retaining the homotopies encoded by families of data. The following tree description can be taken as the concrete meaning of this phrase here.

5.3 The tree model

To construct the morphisms from L_0 to L_1 , consider finite levelled planar rooted trees. Label the complementary regions by branes, with L_0, L_1 on the outside. Label each edge above the root vertex by Floer data for its adjacent pair of branes. All these Hamiltonians are negative on K . To each labelled tree assign the tensor product of the following complexes:

1. a Floer complex at every incoming leaf;
2. a multimorphism complex of $\mathcal{H}_K$ at every non-root vertex;
3. the appropriate disc-chain complex $\mathcal{O}(k)$ at the root.



One summand of the morphism complex from L_0 to L_3 : Floer cochains at the leaves, Floer-data chains at the $\mathcal{H}$ -vertices, and a disk-chain operation at the root.

Take the direct sum over the labelled trees, with the realization shift $[n]$ for a tree with n levels of $\mathcal{H}_K$ -vertices above its $\mathcal{O}$ -root. Thus even the morphisms between a fixed pair can involve intermediate branes. This explains why the canonical complex is much larger than a usual Floer complex.

The differential includes the internal differentials and three types of bar faces. At the leaf level it applies a Floer operation to the incoming cochains. At an intermediate level it composes Floer data. At the root level it forgets the last Floer data using π and composes the resulting disc operations. The alternating signs are those of geometric realization. The relations for these faces, together with the chain-map property of the Floer operations, give $d^2 = 0$.

Finally take the T -adic completion:

$$\mathrm{CF}_M(L_0, L_1; K) = \left(\widehat{\mathbb{L}\pi_*\mathrm{CF}} \right) (L_0, L_1).$$

Here completion allows infinite sums whose valuations tend to $+\infty$. Monotonicity of the Floer data ensures that the differential and operations extend continuously.

To describe the operations, let B_n denote the sum of the tree summands with n levels of $\mathcal{H}_K$ -vertices. Each B_n is already an algebra over $\mathcal{O}_{\mathcal{B}}$: given a composable collection of trees of

the same height and $\theta \in \mathcal{O}(k)$, graft their roots to the k inputs of a new root labelled by θ , and compose θ with the old $\mathcal{O}$ -labels. The result has a single $\mathcal{O}$ -root and still has n levels above it. The bar face maps respect these operations. The degeneracy s_i inserts a whole level of unary identity operations in $\mathcal{H}_K$; it does not change the Floer cochains or the other vertex labels.

Inputs in the realized complex can have different heights $n_1, \dots, n_k$. Put $N = n_1 + \dots + n_k$. A $(n_1, \dots, n_k)$ -shuffle specifies an interleaving of their levels which preserves the order of the levels within each input. For each shuffle, insert identity levels in the j th tree at all the positions assigned to the other trees. All trees now have height N , so the levelwise grafting operation applies. Sum these terms with the shuffle signs and the Koszul signs from the graded tensor factors. This is the Eilenberg–Zilber map followed by the levelwise action:

$$\mathcal{O}(k) \otimes B_{n_1}[n_1] \otimes \dots \otimes B_{n_k}[n_k] \xrightarrow{\text{shuffle}} \mathcal{O}(k) \otimes (B_N^{\otimes k})[N] \xrightarrow{\text{graft}} B_N[N].$$

For example, if both inputs have one level, there are two terms: the nonidentity level of the first tree occurs before that of the second, or after it. For internal degree-zero inputs x, y and a degree-zero binary operation, the shuffle is $s_1x \otimes s_0y - s_0x \otimes s_1y$. These inserted identity levels are simplicial degeneracies of the bar construction, distinct from the cubical parameters in the Floer-data and disc-chain complexes.

The shuffle is a chain map and is compatible with iterated products: iterating shuffles gives precisely the shuffles of all the input levels. Together with associativity of operadic substitution, this makes the realization an algebra over $\mathcal{O}_B$, and valuation non-decreasing operations extend to its completion. Finally, a Boardman–Vogt model chooses compatible fundamental chains for the moduli spaces of discs, using their cubulation by ribbon metric trees. Their boundary formulas define a map of dg operads

$$A_\infty \longrightarrow \mathcal{O}.$$

For $k \geq 2$, evaluating on the chain representing the k -input fundamental class gives μ^k of degree $2 - k$; its boundary is the sum of composites appearing in the A_∞ relation. We use the same choice for every support. See [1, Section 1.1 and Section 6.1].

If $K' \subset K$, negativity on K implies negativity on K' , so $\mathcal{H}_K \subset \mathcal{H}_{K'}$. Inclusion of the corresponding tree summands gives the restriction functor

$$\text{Fuk}_M(K) \longrightarrow \text{Fuk}_M(K').$$

These functors compose strictly in this model. The objects remain the same ambient branes; it is their morphisms that change with the support.

5.4 Why acceleration data still compute the morphisms

The comparison with a telescope is a cofinality statement. In its multicategory form, the relevant condition says that for each fixed composable input tuple $Y_1, \dots, Y_k$, the forgetful maps

$$\mathcal{H}_K(Y_1, \dots, Y_k; X_n) \longrightarrow \mathcal{O}_B(\pi Y_1, \dots, \pi Y_k; \pi X_n)$$

are weak equivalences for all sufficiently large n , where X_n is an acceleration sequence for the output pair. Thus it is not enough merely that the spaces of choices be nonempty. The assertion is that, after the output Hamiltonian is sufficiently large, forgetting the Floer choices loses no homotopical information.

In the Floer construction this follows from the contractibility theorem for the allowed data under a quantitative domination condition on the output Hamiltonian. Negativity on K and

acceleration away from K ensure eventual domination for each fixed finite tuple. Passing from multicategories to categories of finite ordered lists reduces the comparison to the usual cofinality argument for a weighted homotopy colimit. The filtered estimates and completion argument are also needed: an unfiltered comparison alone would not justify passing to completions. The result is a canonical quasi-isomorphism

$$\widehat{\text{tel}}(\text{CF}^*(L_0, L_1; H_n), \kappa_n) \longrightarrow \text{CF}_M(L_0, L_1; K).$$

See [1, Section 7].

This comparison computes the morphism complexes and their cohomology. It does not by itself equip arbitrarily chosen telescopes with compatible products or higher operations. Transferring that structure requires additional chain-level data. We will therefore use telescopes for the homology calculation in the exercise, and the canonical model when stating restriction and descent for A_∞ categories.

6 The exact Liouville picture

6.1 Hamiltonians at infinity and open-string Viterbo restriction

Let (M, λ) be a complete finite-type Liouville manifold, and let L_0, L_1 be exact conical branes as in Section 3. For compact $K \subset M$, choose acceleration Hamiltonians $H_i \rightarrow \chi_K$ which are linear in the radial coordinate sufficiently far out on the cylindrical end. Choose their slopes away from the relevant Reeb-chord lengths and use regular contact-type Floer data. The action-filtered construction of Section 4.2 gives

$$\begin{aligned} \text{CF}_M^*(L_0, L_1; K; \mathbb{Z}) &= \widehat{\text{Tel}}_i^A \text{CF}^*(L_0, L_1; H_i; \mathbb{Z}), \\ \text{HF}_M^*(L_0, L_1; K; \mathbb{Z}) &= H^*(\text{CF}_M^*(L_0, L_1; K; \mathbb{Z})). \end{aligned}$$

Here strips are counted without Novikov weights, and completion uses the action (3). With our Hamiltonian convention, positive radial slopes give negative Reeb flow; identifying chords with $L_0 \cap (\phi_{H_i}^1)^{-1}(L_1)$ describes positive wrapping of the *second* brane.

Now let $D \subset M$ be a compact Liouville subdomain. Assume the branes are conical in a collar of ∂D , meet it transversely, and admit primitives f_i vanishing both on that collar and near infinity. These are standard sufficient compatibility conditions for open-string Viterbo restriction [9, Section 4.1]. Write $L_i^D = \widehat{L_i} \cap \widehat{D}$ for their conical completions. Then the exact comparison gives

$$\text{HF}_M^*(L_0, L_1; D; \mathbb{Z}) \cong \text{HW}_{\widehat{D}}^*(L_0^D, L_1^D; \mathbb{Z}).$$

Thus support on D recovers wrapped morphisms in its completion. The identifications respect the cohomology-level triangle products on these restricted objects.

The proof chooses Hamiltonians adapted to ∂D . The no-escape argument controls strips with the desired boundary conditions, while action separation and completion remove the exterior part of the telescope. The surviving part computes wrapped Floer cohomology in $\widehat{D}$. Its generators have a uniform upper action bound, so completion towards action $+\infty$ adds no infinite sums to this part: the right side is ordinary wrapped Floer cohomology over $\mathbb{Z}$. For nested subdomains with compatible branes, restriction of support becomes the open-string Viterbo map between these wrapped groups. The next exercise gives the elementary action calculation behind this comparison. See [9, Sections 4 and 7] for open-string restriction and its action and no-escape arguments.

6.2 Exercise 1: Viterbo's y -intercept formula

Let $(r_0, \infty) \times Y$ have Liouville form $\lambda = r\alpha$, and let R_α denote the Reeb vector field of α . Take $H = h(r)$ and two exact conical branes L_0, L_1 , with $\lambda|_{L_i} = df_i$. Suppose x is a Hamiltonian chord at radius $r = \rho$, with endpoints on specified conical ends. Set $c_i = f_i(x(i))$.

- (i) Using the convention (1), prove that

$$X_H = -h'(r)R_\alpha.$$

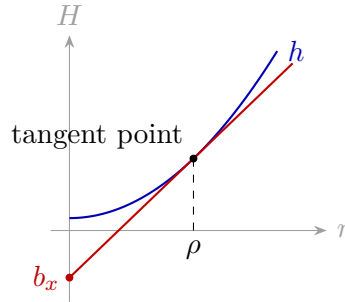
- (ii) Prove the action formula

$$\mathcal{A}_H(x) = h(\rho) - \rho h'(\rho) + c_0 - c_1. \quad (6)$$

Explain why $\mathcal{A}_H(x) - c_0 + c_1$ is the y -intercept of the tangent line to the graph of h at ρ . Include a picture.

- (iii) Show that replacing h by $h + C$ leaves the chords unchanged and raises their actions by C . What is the action on a region where $h(r) = ar + b$?
- (iv) Draw profiles that keep the Hamiltonian fixed near a chosen support while separating the actions of chords in different radial regions. Explain how a family of unwanted chords with actions tending to $+\infty$ can disappear in a completed telescope. Why does merely increasing the slope not by itself imply that the action increases?

Hints. Compute $\iota_{R_\alpha} d(r\alpha)$ and $\lambda(X_H)$. For the last question, differentiate $h(r) - rh'(r)$. The primitives are constant on individual conical ends, but the constants on different ends of a connected brane need not coincide. Use the convergent telescoping sum from Section 2.2.



Identify b_x using the action of the chord.

7 Descent for Poisson-commuting covers

7.1 The geometric condition

Return to closed M and the branes fixed above. A useful sufficient condition on compact subsets $K_1, \dots, K_N$ is that they arise as

$$K_i = F^{-1}(P_i), \quad F = (F_1, \dots, F_m) : M \longrightarrow \mathbb{R}^m, \quad \{F_a, F_b\} = 0,$$

for compact subsets $P_i \subset \mathbb{R}^m$. Functions pulled back from the base then Poisson commute. Preimages of intervals under a torus fibration give the simplest examples.

The preprint uses the following slightly more flexible condition. A finite collection is *weakly Poisson commuting* if there are smooth functions $f_{i,j}$, for $1 \leq i \leq N$ and $j \geq 1$, such that

$$f_{i,j}|_{K_i} < 0, \quad f_{i,j} < f_{i,j+1}, \quad K_i = \bigcap_{j \geq 1} \{f_{i,j} < 0\}, \quad \{f_{i,j}, f_{i',j}\} = 0.$$

A cover $K = \bigcup_i K_i$ of this kind is called *weakly involutive*. All covers in the following theorem are finite and consist of compact sets. This is the precise hypothesis used in [1, Definition 8.1].

7.2 The morphism-level statement

Write $K_I = \bigcap_{i \in I} K_i$ for each nonempty index set I , and abbreviate the canonical supported morphism complex by

$$C_I = CF_M^*(L_0, L_1; K_I; \Lambda_{\geq 0}).$$

The ordered Čech complex is the total complex of

$$\prod_i C_i \longrightarrow \prod_{i < j} C_{ij} \longrightarrow \prod_{i < j < k} C_{ijk} \longrightarrow \cdots.$$

The horizontal maps are alternating sums of restrictions. If an element has internal Floer degree q and Čech degree p , its total degree is $p + q$. One compatible sign convention is $D = \delta_{\check{C}} + (-1)^p d_{\text{Floer}}$.

Theorem 7.1 (Abouzaid–Groman–Varolgüneş). *For a weakly involutive cover of K , the canonical map*

$$CF_M^*(L_0, L_1; K) \longrightarrow \text{Tot } \check{C}^*(\{K_i\}; L_0, L_1)$$

is a quasi-isomorphism for every pair of branes. The totalizations carry compatible A_∞ operations, and restriction defines a quasi-isomorphism

$$\text{Fuk}_M(K) \longrightarrow \check{C}(\text{Fuk}_M; K_1, \dots, K_N)$$

of A_∞ categories with the same brane object set.

This is [1, Theorem 1.4; Section 8]. The result is constructed over $\Lambda_{\geq 0}$ and remains true after extension to Λ .

For two sets, it gives the familiar formula

$$CF_M^*(L_0, L_1; K_1 \cup K_2) \simeq \text{Cone} \left(C_1 \oplus C_2 \xrightarrow{\text{res}_1 - \text{res}_2} C_{12} \right) [-1]$$

and hence the long exact sequence

$$\cdots \rightarrow HF_M^j(K_1 \cup K_2) \rightarrow HF_M^j(K_1) \oplus HF_M^j(K_2) \rightarrow HF_M^j(K_1 \cap K_2) \rightarrow HF_M^{j+1}(K_1 \cup K_2) \rightarrow \cdots,$$

where the brane pair and coefficients have been suppressed.

8 The cylinder inside the two-torus

8.1 The support and the reference meridian

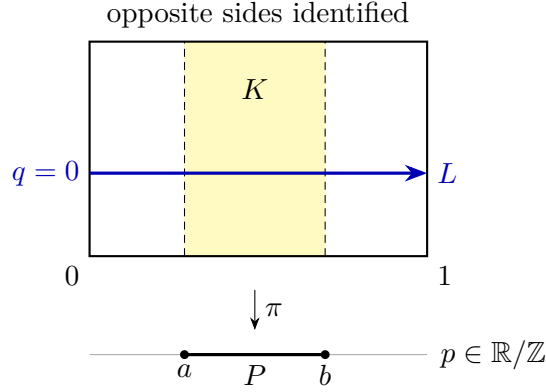
Take the area-one torus

$$M = (\mathbb{R}/\mathbb{Z})_p \times (\mathbb{R}/\mathbb{Z})_q, \quad \omega = dp \wedge dq, \quad \pi(p, q) = p.$$

For a proper closed interval $P = [a, b]$ in the base circle, let

$$K = \pi^{-1}(P), \quad L = (\mathbb{R}/\mathbb{Z})_p \times \{0\}.$$

Thus K is a closed cylinder, while the meridian L is a section of π and is a closed brane in the ambient torus. In Lecture 1 this section was denoted by R . Choose a lift $a < b < a + 1$; the area of the cylinder is $b - a$.



The cut in the q -circle is chosen away from L .

The following exercise is the linear calculation in [1, Section 2.1]. It gives a concrete application of the completed telescope, including the contribution of arbitrarily large winding numbers. The local product needed for mirror gluing is an additional calculation, quoted separately in the next section.

8.2 Exercise 2: supported Floer homology of the meridian

Give L its standard grading and spin structure and the trivial rank-one local system. Work over the Novikov field Λ with valuation val . Compute the supported Floer cohomology $HF_M^*(L, L; K; \Lambda)$, using the completed telescope definition. The goal is to prove an isomorphism of graded Λ -vector spaces

$$HF_M^*(L, L; K; \Lambda) \cong \left\{ \sum_{N \in \mathbb{Z}} a_N \alpha_N : a_N \in \Lambda, \quad \text{val}(a_N) \longrightarrow +\infty \text{ as } |N| \longrightarrow \infty \right\}, \quad |\alpha_N| = 0.$$

In particular, prove vanishing in every other degree. No product computation is requested.

Hints.

- Choose a monotone acceleration datum $H_i = h_i \circ \pi$, with $h_i < 0$ on P , $h_i \rightarrow 0$ on P , and $h_i \rightarrow +\infty$ outside P . Choose the profiles so that for each fixed $N \in \mathbb{Z}$, and all large i , the equation $-h'_i(p) = N$ has exactly two nondegenerate solutions. One way to arrange the profiles is to start with $h_i = ig - 2^{-i}v$, where $g \geq 0$ vanishes exactly on P , has a single nondegenerate maximum on the complementary arc, increases and then decreases on that arc, and is convex just outside the endpoints of P . Take $v > 0$ with $v|_P$ strictly concave and with its maximum in the interior. Smooth the profiles and, if necessary, make small generic adjustments away from the lower chords. Check monotonicity from $h_{i+1} - h_i = g + 2^{-i-1}v > 0$. A chord in class N is $x(t) = (p, Nt)$. Identify its class with its winding in the q -direction. Label the two chords $\alpha_{i,N}, \beta_{i,N}$ so that their degrees are 0, 1; equivalently, they are the index-0 and index-1 zeros of $dh_i + N dp$ on the circle.
- With $J\partial_p = \partial_q$, lift a strip in class N and write $q(s, t) = Nt + r(s, t)$. The Floer equation is

$$p_s = r_t + N + h'_i(p), \quad p_t = -r_s.$$

Show that

$$r_{ss} + r_{tt} - h''_i(p)r_s = 0.$$

Since r vanishes on $t = 0, 1$ and at the two ends, the maximum principle implies $r = 0$. Thus the strips are precisely

$$u(s, t) = (\gamma(s), Nt), \quad \dot{\gamma} = h'_i(\gamma) + N.$$

Draw the two trajectories from $\alpha_{i,N}$ to $\beta_{i,N}$. Their signs are opposite, and their topological energies differ by $|N|$. Deduce, up to multiplying a generator by a unit,

$$d\alpha_{i,N} = T^{A_{i,N}}(1 - T^{|N|})\beta_{i,N}, \quad d\beta_{i,N} = 0,$$

where $A_{i,N} \rightarrow +\infty$ for every fixed N . For $N = 0$ the two contributions cancel. One may work in characteristic two to avoid orientation signs.

(c) Write the continuation maps in class N as

$$\alpha_{i,N} \mapsto u_{i,N} T^{\delta_{i,N}} \alpha_{i+1,N}, \quad \beta_{i,N} \mapsto v_{i,N} T^{\Delta_{i,N}} \beta_{i+1,N},$$

with $u_{i,N}, v_{i,N}$ units of valuation zero. Choose the profiles so that the lower chords remain near K (for $N \neq 0$ they approach its boundary) and the upper chords lie on the high part of the Hamiltonian. Show that

$$\sum_i \delta_{i,N} < \infty, \quad \sum_i \Delta_{i,N} = +\infty.$$

For the lower chords, choose lifts of their p -coordinates near P and consider the tilted critical values $\ell_{i,N} = h_i(p_\alpha) + Np_\alpha$. They tend to $\min(Na, Nb)$. The short continuation trajectory has exponent $\ell_{i+1,N} - \ell_{i,N}$, so its cumulative valuation is bounded. For $N \neq 0$, use the chain-map equation to obtain $\Delta_{i,N} - \delta_{i,N} = A_{i+1,N} - A_{i,N}$. For $N = 0$, use the divergence of the upper critical value directly.

(d) Work first modulo T^E over $R_E = \Lambda_{\geq 0}/\Lambda_{\geq E}$. For fixed N , the differential is zero after discarding finitely many stages. The upper continuation ray has zero direct limit, whereas the lower ray survives. Normalize the surviving generator by its limiting action. Take the inverse limit over E , and only then extend scalars to Λ . Explain why the fact that each finite-stage Floer complex has only finitely many winding classes gives the convergence condition in the answer. In particular, explain why computing finite-stage Floer cohomology over Λ first would lose the answer for $N \neq 0$.

9 From supported categories to the mirror

9.1 A local endomorphism algebra

Let $M = \bigcup_i K_i$ be a finite Poisson-commuting cover, and choose a reference brane L . In this section $\mathrm{Fuk}(M; \Lambda)$ denotes the global supported model $\mathrm{Fuk}_M(M; \Lambda)$. Set

$$\mathcal{A}_I = \mathrm{Hom}_{\mathrm{Fuk}_M(K_I; \Lambda)}(L, L).$$

These are A_∞ algebras with restriction maps. For a global brane L' , the supported morphism complex

$$\mathcal{Y}_I(L') = \mathrm{Hom}_{\mathrm{Fuk}_M(K_I; \Lambda)}(L, L')$$

is a right $\mathcal{A}_I$ -module, with its operations given by precomposition. This is the local version of the restricted Yoneda construction used for T^*S^1 in Lecture 1.

Let $\mathcal{F}_{L,K} \subset \mathrm{Fuk}(M; \Lambda)$ be the full subcategory of global branes which belong to the split-closed category generated by L on every nonempty intersection K_I . Local Yoneda then embeds those split-closed local categories into $\mathrm{Perf}(\mathcal{A}_I)$. Descent gives a cohomologically fully faithful functor

$$\mathcal{F}_{L,K} \longrightarrow \mathrm{holim}_{\emptyset \neq I} \mathrm{Perf}(\mathcal{A}_I). \quad (7)$$

The transition functors here use restriction and the corresponding derived extension of scalars. The homotopy limit retains the identifications and higher compatibility data on intersections.

To obtain a geometric mirror, suppose that the local algebras can be identified, compatibly with restriction, with commutative affinoid algebras A_I , and that their spectra glue by affinoid-domain identifications to a rigid analytic space Y . Analytic descent for perfect complexes then identifies the target of (7) with $\mathrm{Perf}(Y)$. These commutativity and gluing properties are additional geometric inputs; an arbitrary A_∞ algebra does not have an ordinary analytic spectrum.

There are two generation questions. First, do all the desired global Fukaya objects lie in, or become split-generated by, $\mathcal{F}_{L,\mathcal{K}}$? Second, does the image of the resulting functor split-generate $\mathrm{Perf}(Y)$? With affirmative answers, passing to twisted complexes and idempotent summands gives HMS. Descent itself supplies full faithfulness on the indicated subcategory, while these questions supply the required size of source and image.

9.2 Annuli and the Tate curve

For the cylinder $K = \pi^{-1}([a, b])$ in the area-one torus, the additive calculation of Exercise 2 is concentrated in degree zero. A further calculation identifies its cohomology algebra with

$$A_{[a,b]} = \Lambda\langle x, y \rangle / (xy - T^{b-a}). \quad (8)$$

If $x = T^{-a}z$ and $y = T^b z^{-1}$, the corresponding analytic domain is the closed annulus

$$Y_{[a,b]} = \{z \in \mathbb{G}_{m,\Lambda}^{\mathrm{an}} : a \leq \mathrm{val}(z) \leq b\}.$$

Its functions can also be written

$$A_{[a,b]} = \left\{ \sum_{n \in \mathbb{Z}} c_n z^n : \mathrm{val}(c_n) + \min(na, nb) \longrightarrow +\infty \text{ as } |n| \longrightarrow \infty \right\}.$$

The two endpoint inequalities encode convergence on the whole interval. Restriction to a smaller interval is restriction of these analytic functions. Concentration in degree zero also gives algebraic formality of the endomorphism A_∞ algebra after taking a minimal model: only μ^2 can remain for degree reasons.

We use the three abutting closed intervals

$$P_i = [a_i, b_i] = \left[\frac{i}{3} - \frac{1}{6}, \frac{i}{3} + \frac{1}{6} \right], \quad K_i = \pi^{-1}(P_i), \quad i = 0, 1, 2, \quad (9)$$

interpreted cyclically. They cover the base circle, with $P_i \cap P_{i+1} = \{b_i\}$ and no triple intersection. Descent uses a finite compact cover; covering by the interiors is not required. The preimages form a Poisson-commuting cover of M . The overlap algebra is the radius-zero annulus algebra

$$A_{\{b_i\}} = \Lambda\langle W_i, W_i^{-1} \rangle, \quad W_i = T^{-b_i} z.$$

In each cylinder put

$$X_i = T^{-a_i} z, \quad Y_i = T^{b_i} z^{-1}, \quad X_i Y_i = T^{1/3}.$$

On the common boundary, $Y_i = X_{i+1}^{-1}$. These are affinoid-domain gluings along the algebras just displayed. On the universal cover of the base, translation $p \mapsto p + 1$ corresponds to $z \mapsto Tz$. The resulting analytic curve is the Tate curve

$$Y = \mathbb{G}_{m,\Lambda}^{\mathrm{an}} / T^{\mathbb{Z}}.$$

For $\omega = A dp \wedge dq$, the period is T^A , recovering the parameter of the two-torus HMS statement in Lecture 1.

The torus is an example where the required local calculations and generation statements yield the full equivalence

$$D^\pi \text{Fuk}(T^2; \Lambda) \simeq \text{Perf}(Y).$$

As Y is a smooth proper curve, its perfect complexes are the same as its bounded coherent derived category. The present lecture explains the supported local models and the gluing mechanism. The generation inputs and the local product calculation are separate steps in completing this proof.

9.3 The reference section and the fiber circles

The local Yoneda module of the reference section is

$$\mathcal{Y}_I(L) = \mathcal{A}_I.$$

It is the rank-one free module over itself, and its restriction maps are those of the local algebras. Consequently it glues to $\mathcal{O}_Y$. This is a statement built into the construction of the functor, rather than a subsequent identification of its image.

Now take the fiber circle $F_a = \{p = a\} \subset M$, graded so that its unique intersection with L has degree zero. Equip it with a rank-one local system whose holonomy around the positive q -circle is $\xi \in \Lambda^\times$, with $\text{val}(\xi) = 0$. Fix the spin convention in which the triangle counts below have positive sign; changing the spin structure is absorbed by replacing ξ by $-\xi$. For a lifted interval $P = [u, v]$ containing a lift of a , the fiber is contained in $\pi^{-1}(P)$. The support property gives

$$H^* \mathcal{Y}_P(F_a, \xi) = HF_M^*(L, (F_a, \xi); \pi^{-1}(P); \Lambda) \cong \Lambda e, \quad |e| = 0.$$

There is just one intersection generator and its differential vanishes. To determine the module, use the generators $x = T^{-u}z$ and $y = T^v z^{-1}$ of A_P . The triangle counts give

$$e \cdot x = T^{a-u} \xi^{-1} e, \quad e \cdot y = T^{v-a} \xi e.$$

The areas are $a - u$ and $v - a$. For the winding-one input, the local-system map transports from the fiber input to the output in the negative q -direction, which explains ξ^{-1} in this right Yoneda module. The winding-minus-one input uses the opposite transport. Thus

$$e \cdot z = T^a \xi^{-1} e, \quad H^* \mathcal{Y}_P(F_a, \xi) \cong A_P / (z - T^a \xi^{-1}).$$

This is a perfect module, with the two-term free resolution given by multiplication by $z - T^a \xi^{-1}$. If P does not contain a modulo $\mathbb{Z}$, the supported module is zero. These local modules glue to

$$(F_a, \xi) \mapsto \mathcal{O}_{y_{a,\xi}}, \quad y_{a,\xi} = [T^a \xi^{-1}] \in \mathbb{G}_{m,\Lambda}^{\text{an}} / T^{\mathbb{Z}} = Y.$$

Changing the lift a to $a + 1$ multiplies the coordinate by T , so it represents the same point of Y . In particular, a fiber with trivial local system maps to the skyscraper sheaf $\mathcal{O}_{[T^a]}$.

9.4 Two further sections and their mirror line bundles

We use the cover (9). The exact, punctured-torus version in Section 9.5 gives a simpler first example of the polygon and descent calculations below.

The half-translate. Take

$$L_{1/2} = (\mathbb{R}/\mathbb{Z})_p \times \{\tfrac{1}{2}\},$$

with the grading and spin structure obtained by translating those of L , and the trivial local system. On a proper interval P choose a real lift of p . The Hamiltonian $H = -p/2$ translates L to $L_{1/2}$ over P ; choose the lift on a slightly larger interval containing P and extend this Hamiltonian smoothly to the base circle. The corresponding local continuation element is an invertible degree-zero morphism in the supported category. Thus

$$H^*\mathcal{Y}_P(L_{1/2}) \cong A_P e_P, \quad |e_P| = 0.$$

Equivalently, repeat the cylinder calculation with chords of winding $N + \frac{1}{2}$ from L to $L_{1/2}$. The lower generators form one free module over the integer-winding algebra A_P .

Let us calculate the gluing of these frames. On the strip $0 \leq q \leq \frac{1}{2}$ use $\lambda = -q dp$, so that $d\lambda = \omega$ and the primitive on $L_{1/2}$ is $f(p) = -p/2$. Changing the lift from p to $p+1$ changes f by $-1/2$. The area normalization of continuation generators therefore contributes

$$T^{f(p)-f(p+1)} = T^{1/2}.$$

Geometrically, this is the area of the rectangle between the two sections as p travels once around the base. Normalize the frames on the other overlaps so that their transition functions are 1. The remaining identification, over the deck transformation $z \mapsto Tz$, is multiplication by $T^{1/2}$. Consequently the mirror is the explicit line bundle

$$\mathcal{L}_{1/2} = (\mathbb{G}_{m,\Lambda}^{\text{an}} \times \mathbb{A}^1) / ((z, v) \sim (Tz, T^{1/2}v)) \longrightarrow Y.$$

Thus a section upstairs is a function satisfying $f(Tz) = T^{1/2}f(z)$. The bundle has degree zero and is nontrivial: comparison of Laurent coefficients shows that this equation has no nonzero global analytic solution, since every Laurent exponent is an integer. Its square is trivial, since the squared multiplier is T and z is a nowhere-vanishing solution of $f(Tz) = Tf(z)$.

A section with three intersections: a direct Floer computation. Take

$$L' = \{q = 3p\} \subset M, \quad x_i = (i/3, 0), \quad i = 0, 1, 2,$$

with the section grading, so that the three intersections have degree zero, and with trivial local system. Choose compatible spin conventions for the positive counts below. Let

$$P_i = [i/3 - 1/6, i/3 + 1/6], \quad K_i = \pi^{-1}(P_i),$$

where intervals are interpreted cyclically. Each intersection lies in just its own cylinder, away from all overlaps. The local arcs are Hamiltonian isotopic to those of L . We will prove directly that

$$e_i := \text{res}_{K_i}(x_i)$$

is a generator of the local module, and compute the transitions using continuation strips and triangles. No evaluation on fiber objects is used.

The winding coordinate. For a lifted interval $P = [a, b]$, put

$$A_P = HF_{\pi^{-1}(P)}^0(L, L; \Lambda), \quad s_P(k) = \min_{p \in P} kp.$$

Write α_k for the lower surviving winding- k class from Exercise 2. Fix compatible signs and scalar normalizations so that

$$\alpha_k \alpha_\ell = T^{s_P(k+\ell) - s_P(k) - s_P(\ell)} \alpha_{k+\ell}.$$

The coordinate used below is the element of this Floer algebra

$$z = T^a \alpha_1, \quad z^{-1} = T^{-b} \alpha_{-1}, \quad z^k = T^{s_P(k)} \alpha_k.$$

Thus z is an action-normalized winding-one class. Series in z are series in the completed Floer algebra; their convergence means $\text{val}(c_k) + s_P(k) \rightarrow +\infty$ for $\sum_k c_k z^k$.

The module generators and the continuation count. Define

$$f_N(p) = \frac{3}{2}p^2 - Np, \quad m_P(N) = \min_{p \in P} f_N(p).$$

For an acceleration Hamiltonian $h(p)$, a chord from L to L' has

$$q(t) = (3p - N)t, \quad h'(p) + 3p - N = 0.$$

The deck identification is $(p, N) \mapsto (p + 1, N + 3)$. To compute strips, on the universal cover use the shear

$$(p, r) \mapsto (p, r + t(3p - N))$$

and the almost complex structure obtained by pushing forward the standard one. Its derivative is independent of N and of the point, so the resulting J_t descends to the torus. Both boundary conditions become $r = 0$. For continuation data h_s , the equations become

$$p_s = r_t + h'_s(p) + 3p - N, \quad p_t = -r_s.$$

Consequently $r_{ss} + r_{tt} - (h''_s(p) + 3)r_s = 0$; the maximum principle gives $r = 0$. The strips are exactly the solutions of

$$p_s = \partial_p G_{s,N}(p), \quad G_{s,N}(p) = h_s(p) + f_N(p).$$

This also checks the grading: the x_i are the minima of f_i .

Start with $h = 0$ and consider the component represented by $N = i$. For each minimum c of the final function $G_i = h + f_i$, there is exactly one continuation strip from x_i to c . Indeed, where the homotopy is already constant near $+\infty$, convergence to this minimum forces the solution to be constant at c . Integrate backward uniquely. Near $-\infty$ the solution is $p(s) = i/3 + Ae^{3s}$, hence converges to x_i . Its topological energy is

$$G_i(c) - f_i(i/3) = G_i(c) + i^2/6.$$

Small constant shifts arrange the strict negativity and monotonicity required by acceleration; they cancel in the action normalization.

As h accelerates, a lower minimum survives near each lifted interval $P + n$. Translate it back to P and label it by $N = i - 3n$. Let β_N be its limiting lower-chord class, normalized as in Exercise 2. The limiting continuation exponent is

$$m_P(N) + N^2/6 = \min_{p \in P} \frac{3}{2}(p - N/3)^2.$$

Upper critical points lie on the diverging barriers and disappear in the completed telescope. The direct continuation count therefore gives

$$\boxed{\text{res}_P(x_i) = \sum_{N \equiv i \pmod{3}} T^{m_P(N) + N^2/6} \beta_N.} \quad (10)$$

The coefficients tend to arbitrarily high valuation as $|N| \rightarrow \infty$.

The triangle count. We next compute the right action of α_k on β_N . For large acceleration, work in one lift of the cylinder and choose convex lower profiles. The relevant three graph arcs can be written

$$C_0 : q = 0, \quad C_1 : q = F_1'(p), \quad C_2 : q = (F_1 + F_2)'(p),$$

where

$$F_1 = h + kp, \quad F_2 = h + \frac{3}{2}p^2 - Np.$$

Their intersections are the minima of F_1 , F_2 , and $F_1 + F_2$; the last lies between the first two. The arcs bound one embedded triangle, with count $+1$. Stokes' theorem gives its area as

$$\min(F_1 + F_2) - \min F_1 - \min F_2.$$

A small perturbation treats coinciding minima. Additional triangles leaving this lift cross a high lobe of an acceleration graph. Their areas tend to $+\infty$, so they contribute zero modulo T^E once the acceleration is sufficiently large. Taking this limit first for each E gives

$$\boxed{\beta_N \alpha_k = T^{m_P(N-k) - m_P(N) - s_P(k)} \beta_{N-k}.} \quad (11)$$

The exponent is nonnegative because $f_{N-k} = f_N + kp$. Normalize the explicit chord classes by

$$\gamma_N = T^{m_P(N)} \beta_N.$$

Then (11) says simply

$$\gamma_N z^k = \gamma_{N-k}.$$

The coefficient norms in this description agree with the completed module norms up to bounded factors, since

$$\min_P \frac{3}{2}p^2 \leq m_P(N) - s_P(-N) \leq \max_P \frac{3}{2}p^2.$$

Thus γ_0 generates the completed local module. This auxiliary notation refers to a specified lower chord and its action normalization.

Transitions using the intersection classes themselves. Combining the two counts gives, inside the completed Floer module,

$$e_i = \gamma_0 \Theta_i(z), \quad \Theta_i(z) = \sum_{N \equiv i \pmod{3}} T^{N^2/6} z^{-N} = T^{i^2/6} z^{-i} U_i(z),$$

where

$$U_i(z) = \sum_{n \in \mathbb{Z}} T^{3n^2/2 - in} z^{3n}.$$

In the winding generators $X_i = T^{-a_i} z$ and $Y_i = T^{b_i} z^{-1}$, the same series is

$$U_i = 1 + \sum_{n \geq 1} T^{n(3n-1)/2} (X_i^{3n} + Y_i^{3n}). \quad (12)$$

Thus $U_i = 1 + V_i$ with V_i of strictly positive filtration, and

$$U_i^{-1} = \sum_{\ell \geq 0} (-V_i)^\ell$$

converges in A_{P_i} . Hence $e_i = \text{res}(x_i)$ is indeed a local generator.

On an overlap, define g_{ij} by $e_i = \mu^2(e_j, g_{ij})$. The already-computed triangle action gives

$$\boxed{g_{ij} = T^{(i^2 - j^2)/6} z^{j-i} U_i U_j^{-1}.} \quad (13)$$

In cyclic order,

$$g_{01} = T^{-1/6} z U_0 U_1^{-1}, \quad g_{12} = T^{-1/2} z U_1 U_2^{-1}, \quad g_{20} = T^{-5/6} z U_2 U_3^{-1}.$$

The last expression uses $P_3 = P_0 + 1$. Reindexing gives $\Theta_3 = \Theta_0$, which identifies x_3 with x_0 . Equivalently, on each cyclic overlap the formula is

$$e_i = X_{i+1} \frac{U_i}{U_{i+1}} e_{i+1}, \quad Y_i X_{i+1} = 1. \quad (14)$$

Every inverse and product here is taken in the completed winding algebra; the formulas do not require evaluation at points.

The modules therefore glue to a line bundle $\mathcal{E}$. The global Floer complex has the three degree-zero generators x_0, x_1, x_2 and zero differential. Descent gives

$$R\Gamma(Y, \mathcal{E}) \simeq \Lambda^3[0], \quad h^0(Y, \mathcal{E}) = 3, \quad h^1(Y, \mathcal{E}) = 0.$$

Riemann–Roch then gives $\deg \mathcal{E} = 3$. Thus the three intersection classes give a basis of global sections, and their restrictions give the local trivializations used in (13).

The area factors for a general section. The same calculation applies to a lift $L' : q = f(p)$ with $f(p+1) = f(p) + 3$ and precisely three degree-zero crossings. Choose $f(c_i) = i$, $c_{i+3} = c_i + 1$, and

$$F' = f, \quad F(p+1) = F(p) + 3p + C, \quad u_i = F(c_i) - i c_i.$$

The strip reduction now uses $h + F - Np$, and the same unique backward continuation trajectories. With $m_P(N) = \min_P(F - Np)$ and $\gamma_N = T^{m_P(N)} \beta_N$, it gives

$$\text{res}_P(x_i) = \gamma_0 T^{-u_i} z^{-i} U_i(z), \quad U_i(z) = \sum_{n \in \mathbb{Z}} T^{\frac{3}{2}n(n-1) + (C-i)n} z^{3n}.$$

Whenever these restrictions are generators on the chosen cylinders,

$$g_{ij} = T^{u_j - u_i} z^{j-i} U_i U_j^{-1}.$$

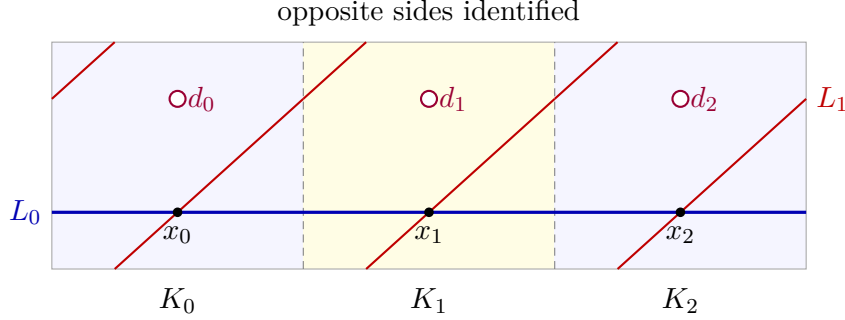
A sufficient condition for the geometric-series inversion is $P_i \subset ((i - C)/3, (i + 3 - C)/3)$. This spells out the nonvanishing condition on the chosen intersection trivializations; local Hamiltonian isotopy alone establishes freeness of the module.

9.5 The large-volume limit: direct descent in the exact case

Here is a simpler version of the preceding computation, using only unweighted polygons in an exact surface. Keep the three abutting cylinders (9), and put

$$d_i = (i/3, 1/2), \quad D = \{d_0, d_1, d_2\}, \quad L_0 = \{q = 0\}, \quad L_1 = \{q = 3p\}.$$

There is one puncture in each of the three complementary disks of $L_0 \cup L_1$. The intersections are still $x_i = (i/3, 0)$, all of degree zero. We compute over a field $\mathbb{k}$; choose the same spin and sign conventions as above, or work in characteristic two.



A fundamental domain, cut at $q = -1/4, 3/4$. The open circles are the deleted points.

The exact geometry. Choose a $p \mapsto p + 1/3$ -invariant primitive of ω on $T^2 \setminus D$ with equal positive residues $1/3$. Adding a closed form $A dp + B dq$ makes both L_0 and L_1 exact, since their homology classes $(1, 0)$ and $(1, 3)$ are independent. The common circles $p = b_i$ are then exact as well: the difference between the L_1 and L_0 arcs from x_i to x_{i+1} differs from this circle by the boundaries of two oppositely oriented triangles of equal area $1/24$, avoiding D . Translation invariance and exactness make the integrals along both arcs zero.

An exact change of primitive therefore arranges $\lambda = (p - b_i)dq$ near each common circle. Slightly thicken the abutting cylinders in these collars, remove small disks around the d_i , and make the branes conical near the new boundaries. This gives Liouville pairs of pants V_i with Liouville annuli $V_{i,i+1}$ as overlaps, covering a Liouville core. The collar cutoffs can be chosen to depend only on p , so this is a Poisson-commuting cover. Use the completions for the local wrapped calculations. This also replaces the finite-area puncture ends by complete Liouville ends. The symbols K_i specify the decomposition; the V_i provide its Liouville realization with collars.

The local algebra, computed by triangles. Let ℓ_i be the completed reference arc $L_0 \cap V_i$. After wrapping, its endomorphism generators are

$$1, \quad X_i^m, \quad Y_i^n \quad (m, n \geq 1).$$

Here X_i goes once around the left cylinder end and Y_i once around the right end. All generators have degree zero, so the differential vanishes. In a collar, lifting the arcs gives exactly one triangle concatenating two windings at the same end. Thus

$$X_i^m X_i^n = X_i^{m+n}, \quad Y_i^m Y_i^n = Y_i^{m+n}.$$

There is no mixed triangle. To check this topologically, collapse the reference arc to a basepoint. The two end loops freely generate the fundamental group of the pair of pants. A mixed word cannot be the pure power, or the identity, represented by an output chord. Equivalently, the candidate triangle in the filled cylinder encloses the removed point d_i . Consequently

$$A_i^0 = HW^*(\ell_i, \ell_i; \mathbb{k}) = \mathbb{k}[X_i, Y_i]/(X_i Y_i), \quad |X_i| = |Y_i| = 0. \quad (15)$$

These are ordinary polynomials: the exact wrapped direct limit has finite sums, with no Novikov completion. The same surface computation is described in [7, Section 3].

Restrictions to the overlap. Write s_i for positive winding in the common annulus, with the orientation chosen so that it agrees with Y_i . Its algebra is $B_i^0 = \mathbb{k}[s_i, s_i^{-1}]$. Continuation preserves the surviving end chords; the chords at the other end disappear under restriction. The maps are therefore

	first generator	second generator
$A_i^0 \rightarrow B_i^0$	$X_i \mapsto 0$	$Y_i \mapsto s_i$
$A_{i+1}^0 \rightarrow B_i^0$	$X_{i+1} \mapsto s_i^{-1}$	$Y_{i+1} \mapsto 0$.

These are the localizations at Y_i and X_{i+1} , respectively. Thus $\text{Spec } A_i^0$ is a node chart, and its right branch glues to the left branch of the next chart by $Y_i X_{i+1} = 1$. We obtain a curve Y_0 consisting of three $\mathbb{P}^1$'s in a cycle. This is already an algebraic descent calculation in the exact setting.

The section module and its transitions, computed directly. Let ℓ'_i be the completed arc from L_1 . Its unwrapped intersection with ℓ_i is $e_i = \text{res}_{V_i}(x_i)$. The remaining generators are $e_i X_i^m$ and $e_i Y_i^n$: the unique triangles in the two end collars give these actions. The same mixed-word obstruction excludes the other products. Hence

$$M_i^0 = HW^*(\ell_i, \ell'_i; \mathbb{k}) = e_i A_i^0.$$

In particular, the specified intersection class itself is a basis.

Unroll the punctured torus as in the picture. The short continuation strips from the neighboring intersections to this cylinder avoid the punctures. The left one contributes $X_i e_i$, the right one $Y_i e_i$, and the strip from x_i contributes e_i . Each count is 1. Any additional winding adds a positive polygon domain containing one of the punctured complementary disks, so gives no strip in the exact surface. We have directly obtained

$$(\text{res}_{V_i} x_{i-1}, \text{res}_{V_i} x_i, \text{res}_{V_i} x_{i+1}) = (X_i, 1, Y_i) e_i. \quad (16)$$

On the common annulus, the two restrictions of x_{i+1} agree. Alternatively, its adjacent winding generators are related by the unique annular triangle. Either description gives

$$e_{i+1} = Y_i e_i, \quad \boxed{e_i = X_{i+1} e_{i+1}}. \quad (17)$$

No theta series or evaluation on fiber objects enters this calculation.

The reference modules glue to $\mathcal{O}_{Y_0}$. The modules M_i^0 glue to a line bundle $\mathcal{E}_0$. On the component with affine coordinates $s_i = Y_i$ and $s_i^{-1} = X_{i+1}$, (17) is the transition for $\mathcal{O}_{\mathbb{P}^1}(1)$. Thus

$$\deg(\mathcal{E}_0|_{C_i}) = 1 \quad (i = 0, 1, 2).$$

Moreover, (16) identifies the sections with the three homogeneous coordinates of

$$Y_0 = \{S_0 S_1 S_2 = 0\} \subset \mathbb{P}^2, \quad \mathcal{E}_0 = \mathcal{O}_{\mathbb{P}^2}(1)|_{Y_0}.$$

In particular the descended line bundle is ample.

Checking descent on the global morphism group. There are no triple overlaps, so the descent complex for (L_0, L_1) is

$$\bigoplus_i A_i^0 e_i \xrightarrow{\delta} \bigoplus_i B_i^0 e_{i+1},$$

in degrees 0, 1. Write a local coefficient as

$$h_i = c_i + f_i(X_i) + g_i(Y_i), \quad f_i(0) = g_i(0) = 0.$$

On overlap i , the compatibility equation is

$$c_i + g_i(s_i) = s_i(c_{i+1} + f_{i+1}(s_{i+1}^{-1})).$$

Comparing Laurent coefficients gives

$$h_i = c_i + c_{i-1}X_i + c_{i+1}Y_i.$$

Thus the kernel is $\mathbb{k}^3$, with precisely the three intersection classes as basis. The map δ is onto: $Y_i^n e_i$, $n \geq 1$, produces $s_i^{n-1} e_{i+1}$ on overlap i and zero on the other overlap, while $X_{i+1}^n e_{i+1}$ produces the negative powers there, up to the sign in δ . Hence

$$H^*(\check{C}(\{V_i\}; L_0, L_1)) = \mathbb{k}^3[0] = HF^*(L_0, L_1; \mathbb{k}).$$

For (L_0, L_0) the same calculation has no factor s_i in the compatibility equation. All nonconstant Laurent terms are in the image; constants give the cochain complex of a circle. It yields $\mathbb{k}$ in degrees 0 and 1, agreeing with $H^*(L_0; \mathbb{k})$.

The three formal parameters. To recover the family in the large-volume picture, allow polygons in the filled torus and give a polygon u the weight

$$Q_0^{u \cdot d_0} Q_1^{u \cdot d_1} Q_2^{u \cdot d_2}, \quad R = \mathbb{k}[[Q_0, Q_1, Q_2]].$$

The previously excluded elementary mixed triangle in cylinder i contains d_i once and no other marked point. It now contributes Q_i times the unit. Same-end triangles still have weight 1. Thus the local relative algebra is, directly,

$$A_i^R = R\langle X_i, Y_i \rangle / (X_i Y_i - Q_i). \quad (18)$$

For example, cancellation of m opposite windings contributes Q_i^m . The brackets mean the (Q_0, Q_1, Q_2) -adic completion of polynomials; modulo each power of that ideal only finitely many monomials occur. In particular the central fiber is the polynomial node algebra (15). The formal overlap is $R\langle s_i, s_i^{-1} \rangle$, with maps

$$(X_i, Y_i) = (Q_i s_i^{-1}, s_i), \quad (X_{i+1}, Y_{i+1}) = (s_i^{-1}, Q_{i+1} s_i).$$

These charts glue to the formal three-gon smoothing over $\mathrm{Spf} R$. The multiplicative period is $Q_0 Q_1 Q_2$. The monomial transitions X_{i+1} define a formal line bundle extending $\mathcal{E}_0$; it is relatively ample because its degree on each central component is 1. For the relative Fukaya category and the corresponding three-parameter Tate family, see [8].

Comparison with the area-weighted calculation. Specialize Q_i to $t = T^{1/3}$. The formal charts above then have $X_i Y_i = t$, and their Novikov generic fibers are exactly the annuli from Section 9.2. Their period is $t^3 = T$. The intersection frames in Section 9.4 have the transitions (14). Formula (12) shows that $U_i \equiv 1 \pmod{t^3}$ on its own chart, so these transitions extend over the formal model and reduce to the directly computed (17). Dividing the local frames by these explicit units identifies this extension with the monomial line bundle just constructed.

The global sections agree as well. On chart i the neighboring coefficients are $X_i U_{i-1}/U_i$ and $Y_i U_{i+1}/U_i$. Here $U_{i-1} \equiv 1 + Y_i^3$ and $U_{i+1} \equiv 1 + X_i^3 \pmod{t}$; multiplication by X_i and Y_i and the relation $X_i Y_i = t$ reduce them to X_i and Y_i . This recovers (16). The exact computation was obtained by counting puncture-avoiding polygons over $\mathbb{k}$; this comparison explains its role as the large-volume central fiber. Reduction is taken in the formal model, before passing to the Novikov field.

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